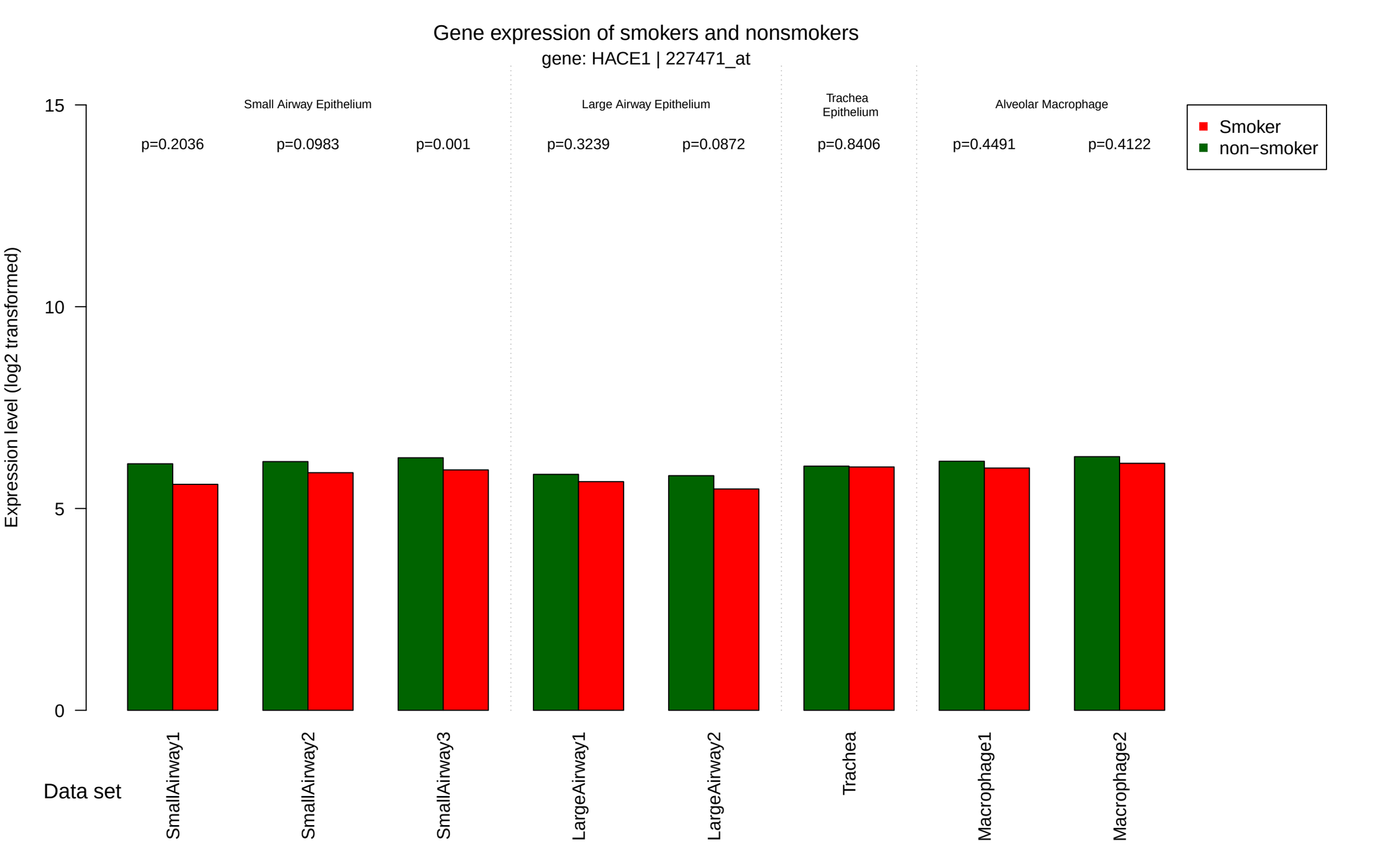
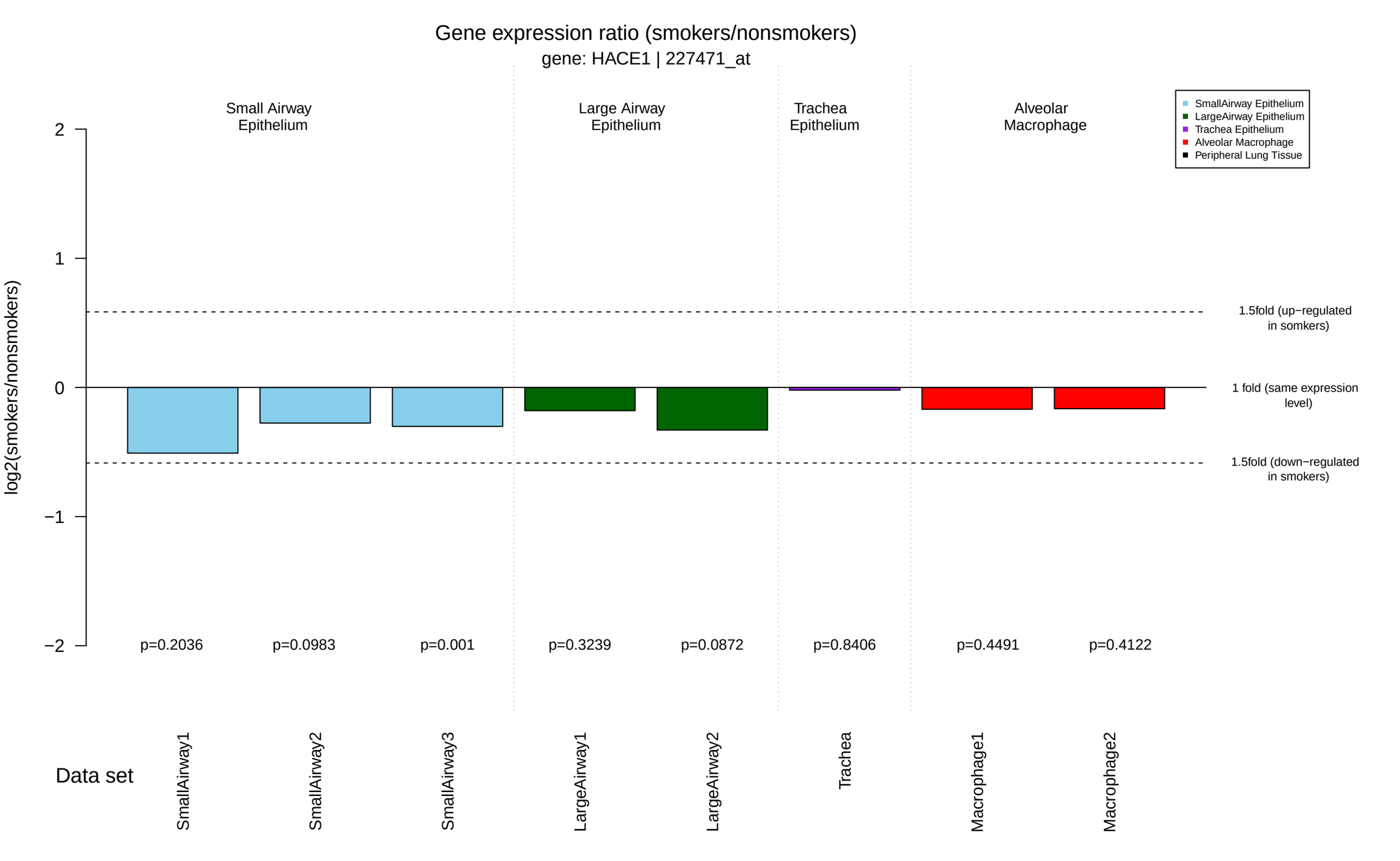
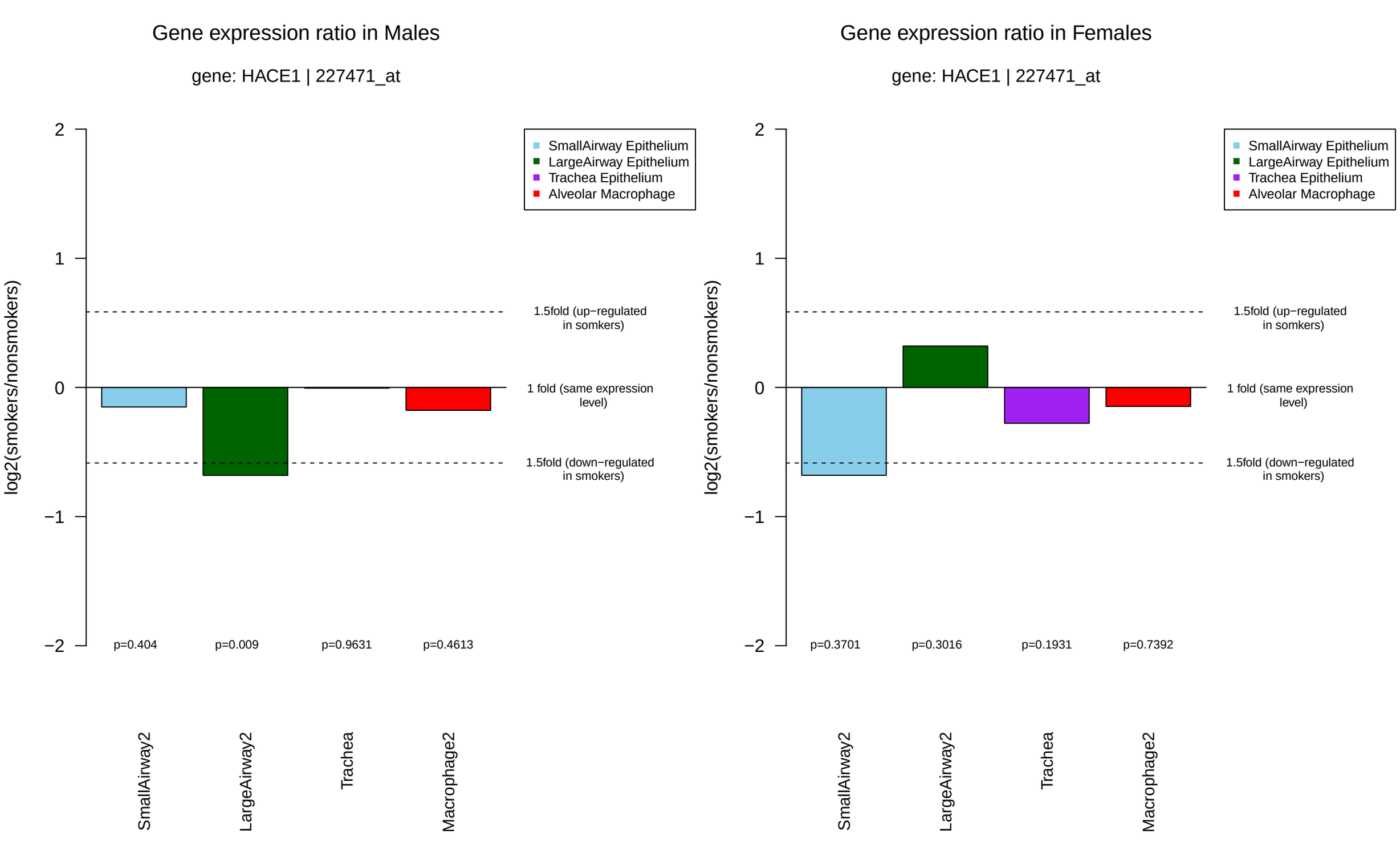
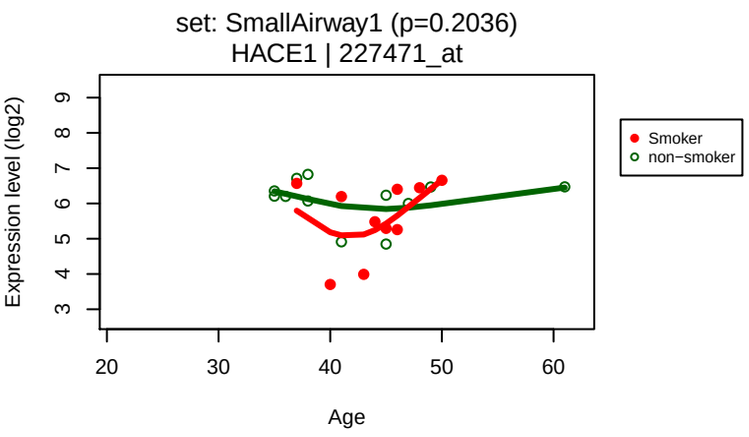
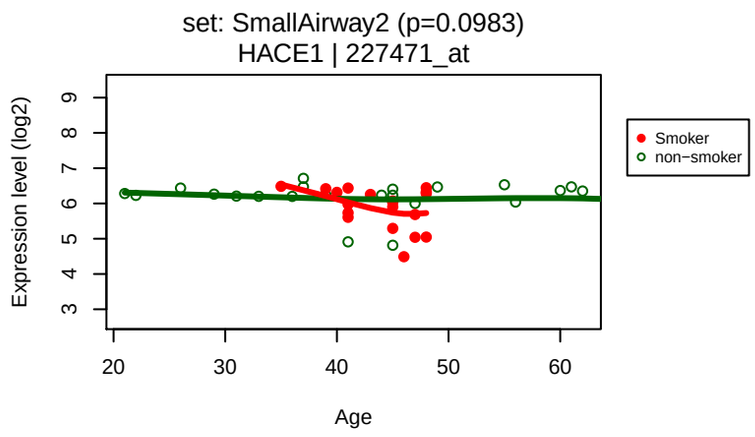
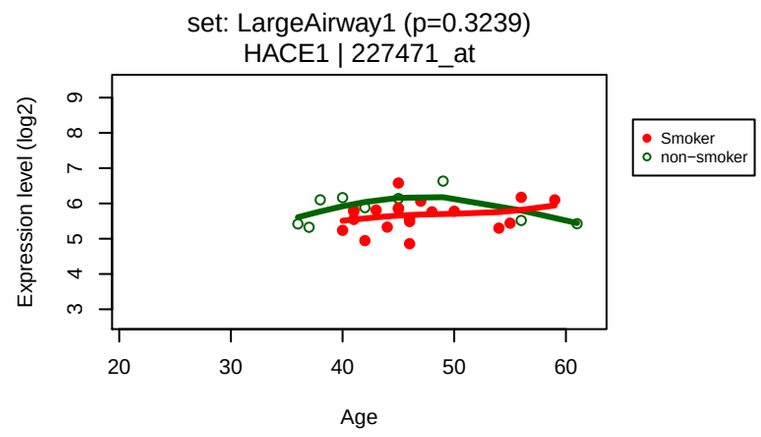

Gene expression of smokers and nonsmokers
gene: HACE1 | 227471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2036
p=0.0983
p=0.001
p=0.3239
p=0.0872
p=0.8406
p=0.4491
p=0.4122
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

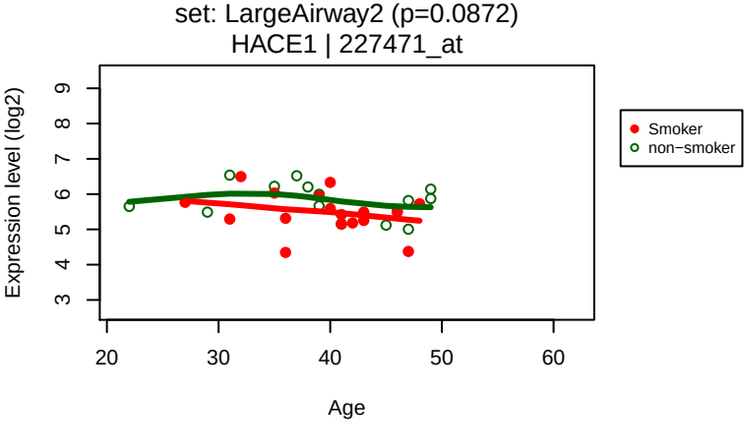

Gene expression ratio (smokers/nonsmokers)
gene: HACE1 | 227471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2036
p=0.0983
p=0.001
p=0.3239
p=0.0872
p=0.8406
p=0.4491
p=0.4122
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

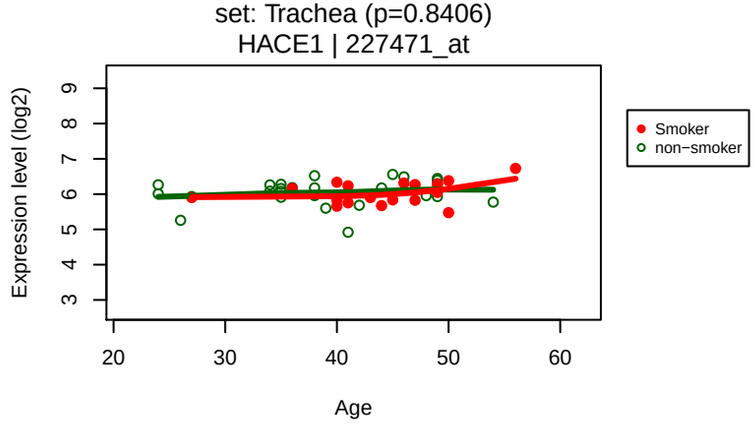

Gene expression ratio in Males
gene: HACE1 | 227471_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.404
p=0.009
p=0.9631
p=0.4613
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HACE1 | 227471_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3701
p=0.3016
p=0.1931
p=0.7392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

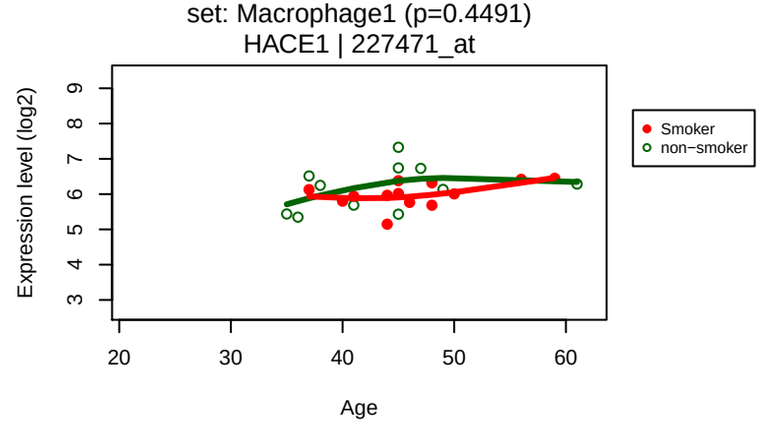

set: SmallAirway1 (p=0.2036)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker

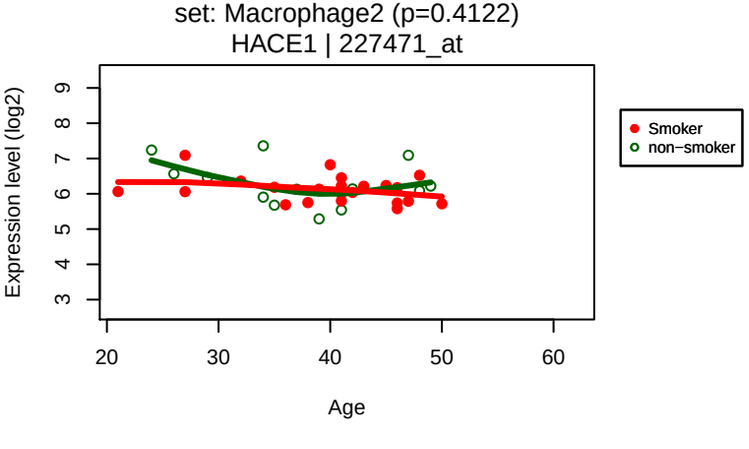

set: SmallAirway2 (p=0.0983)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker

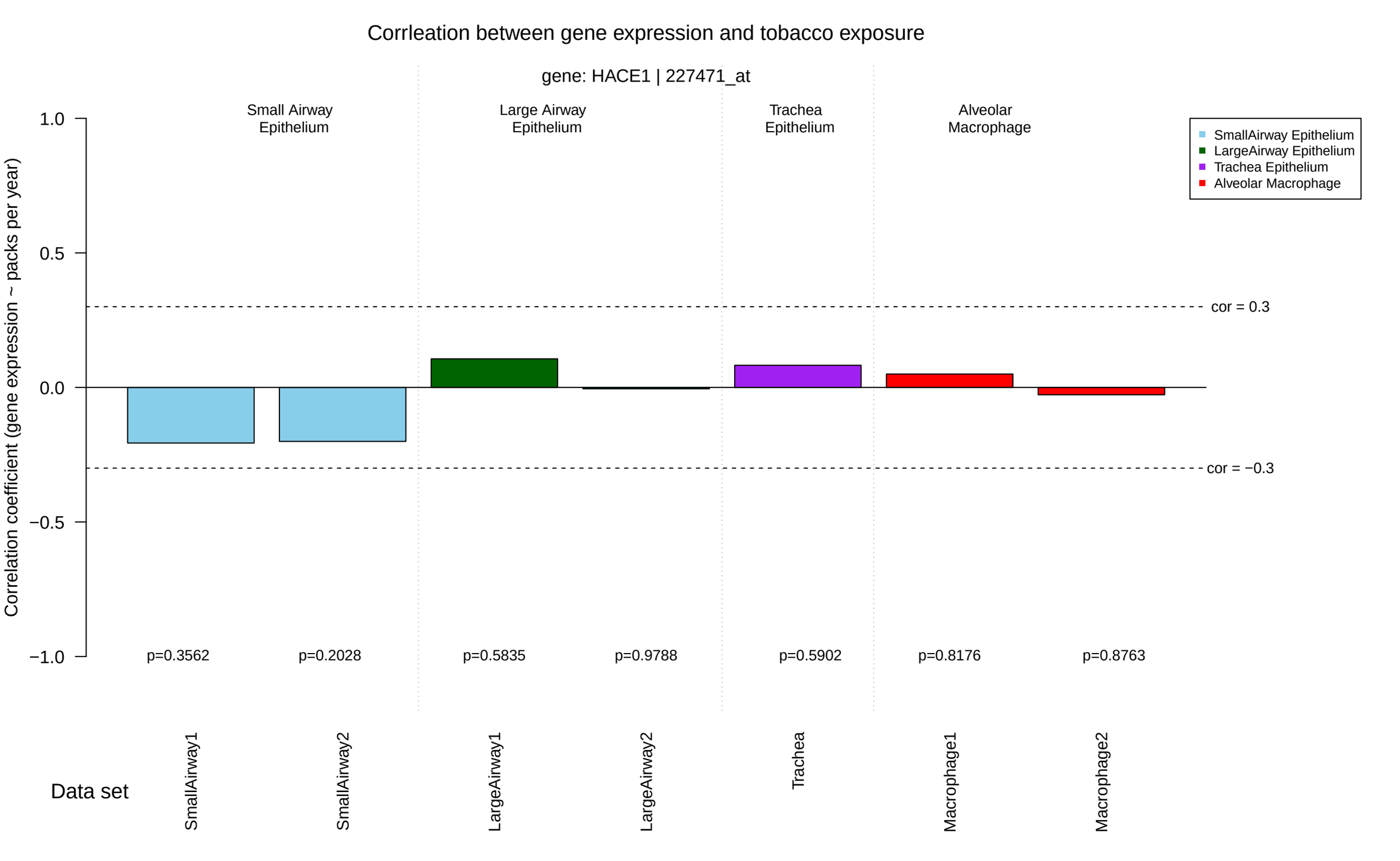

set: LargeAirway1 (p=0.3239)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0872)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8406)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4491)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4122)
HACE1 | 227471_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HACE1 | 227471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3562
p=0.2028
p=0.5835
p=0.9788
p=0.5902
p=0.8176
p=0.8763
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2